

# FORECASTING RESULTS REPORT

Report Date: 20-08-2026  
Forecast Target: Monthly Total Credit  
Test Month: June 2026  
Actual June Credit: █59,000.00

## STAGE 2A — TRADITIONAL FORECASTING MODELS

| Model                  | Forecast   | MAE       | RMSE      | MAPE    |
|------------------------|------------|-----------|-----------|---------|
| ETS (Holt's Double)    | █59,019.84 | █19.84    | █19.84    | 0.0336% |
| SARIMA(1,1,1)(1,1,1,2) | █57,500.00 | █1,500.00 | █1,500.00 | 2.5424% |
| ARIMA(1,1,1)           | █56,900.91 | █2,099.09 | █2,099.09 | 3.5578% |

## STAGE 2B — NEURAL FORECASTING MODELS

| Model      | Forecast   | MAE       | RMSE      | MAPE    |
|------------|------------|-----------|-----------|---------|
| LSTM       | █59,008.61 | █8.61     | █8.61     | 0.0146% |
| GRU        | █57,619.36 | █1,380.64 | █1,380.64 | 2.3401% |
| Simple RNN | █57,026.42 | █1,973.58 | █1,973.58 | 3.3450% |

## FINAL MODEL RANKING

| Rank | Model                  | Type        | Forecast   | MAPE    |
|------|------------------------|-------------|------------|---------|
| 1    | LSTM                   | Neural      | █59,008.61 | 0.0146% |
| 2    | ETS (Holt's Double)    | Traditional | █59,019.84 | 0.0336% |
| 3    | GRU                    | Neural      | █57,619.36 | 2.3401% |
| 4    | SARIMA(1,1,1)(1,1,1,2) | Traditional | █57,500.00 | 2.5424% |
| 5    | Simple RNN             | Neural      | █57,026.42 | 3.3450% |
| 6    | ARIMA(1,1,1)           | Traditional | █56,900.91 | 3.5578% |

## BEST PERFORMING MODEL

Model: LSTM  
Type: Neural  
Forecast: █59,008.61  
Actual: █59,000.00  
MAE: █8.61  
RMSE: █8.61  
MAPE: 0.0146%

## FORECASTING SETUP

Total Months: 6  
Training Months: 5

**Testing Months:** 1  
**Test Observation:** June 2026  
**Actual June Credit:** ■59,000.00

## EVALUATION METRICS

**MAE** — Mean Absolute Error  
**RMSE** — Root Mean Squared Error  
**MAPE** — Mean Absolute Percentage Error

Lower values indicate better forecasting performance.

**Note:** The current experiment contains one held-out test observation, so the metrics represent performance on this particular test split.

## OVERALL TRANSACTION SURPLUS

**Total Credit:** ■340,000.00  
**Total Debit:** ■252,244.00  
**Net Surplus:** ■87,756.00  
**Surplus / Profit %:** 25.81%

**Calculation:**  
 $(\text{Total Credit} - \text{Total Debit}) / \text{Total Credit} \times 100$